

Formation pathway of high-efficiency kesterite solar cells fabricated through molecular ink chemistry

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Alex Jimenez-Arguijo^{1,2,8}✉, Yuancai Gong^{1,2,8}✉, Ivan Caño^{1,2}, Outman El Khouja^{1,2}, Jianjun Li³✉, Kaiwen Sun⁴✉, Zacharie Jehl Li-Kao^{1,2}, Sergio Giraldo^{1,2}, Hao Xin⁵✉, Alejandro Perez-Rodriguez^{6,7}✉, Xiaojing Hao⁴✉ & Edgardo Saucedo^{1,2}✉

Solar cells based on kesterite materials, $\text{Cu}_2\text{ZnSn}(\text{S,Se})_4$ (CZTSSe), offer a non-toxic, Earth-abundant solution for energy generation. However, they have historically struggled to achieve power conversion efficiencies comparable to those of other thin-film photovoltaic technologies. Here we highlight the critical role of the synthesis and formation pathway of these multinary semiconductors, discussing the challenges associated with kesterite layer fabrication and their impact on device performance. In particular, we discuss how the design of molecular inks in kesterite synthesis is key to overcoming these limitations, unveiling the connections between precursor chemistry, synthesis pathways and the formation of point and extended defects. We discuss how precise control over these factors has enabled kesterite solar cells to exceed 15% efficiency. Building on these advances, we propose strategies to further improve device performance. Finally, the insights presented here provide a framework for the exploration and development of other multinary semiconductor materials.

Kesterite, $\text{Cu}_2\text{ZnSn}(\text{S,Se})_4$ (CZTSSe), materials have garnered substantial attention as candidates for sustainable photovoltaics, offering a unique combination of Earth abundance, non-toxicity and compatibility with scalable thin-film processing. However, their power conversion efficiencies (PCEs) have historically fallen behind those of other thin-film technologies.

The multinary nature of CZTSSe compounds, with Cu, Zn and Sn cations, introduces a complex chemical landscape, multiple oxidation

states and a high density of potential intrinsic defects. These defects—ranging from secondary phases to grain boundary and point-level imperfections—are rooted in the complicated formation pathways of CZTSSe compounds. The wide variety of intermediate phases, including volatile species such as $\text{Sn}-(\text{S,Se})$ and highly stable compounds such as $\text{Zn}(\text{S,Se})$, hinder compositional uniformity during synthesis, leading to non-uniform reactions and defect formation. Initially adopted synthesis methods, such as physical vapour deposition (PVD) and

¹Photovoltaic Lab, Micro and Nano Technologies Group, Electronic Engineering Department, Barcelona East School of Engineering, Universitat Politècnica de Catalunya, Barcelona, Spain. ²Barcelona Research Center for Multiscale Science and Engineering, Universitat Politècnica de Catalunya, Barcelona, Spain. ³Shenyang National Laboratory for Materials Science, Institute of Metal Research, Chinese Academy of Sciences, Shenyang, China. ⁴Australian Centre for Advanced Photovoltaics, School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Sydney, New South Wales, Australia. ⁵State Key Laboratory for Organic Electronics and Information Displays, College of Chemistry and Life Sciences, Nanjing University of Posts and Telecommunications, Nanjing, China. ⁶Solar Energy Materials and Systems Group, Catalonia Institute for Energy Research, Sant Adrià de Besòs, Spain. ⁷IN2UB, Departament d'Enginyeria Electrònica i Biomèdica, Universitat de Barcelona, Barcelona, Spain. ⁸These authors contributed equally: Alex Jimenez-Arguijo, Yuancai Gong. ✉e-mail: alex.jimenez.arguijo@upc.edu; yuancai.gong@upc.edu; xj.hao@unsw.edu.au; edgardo.saucedo@upc.edu

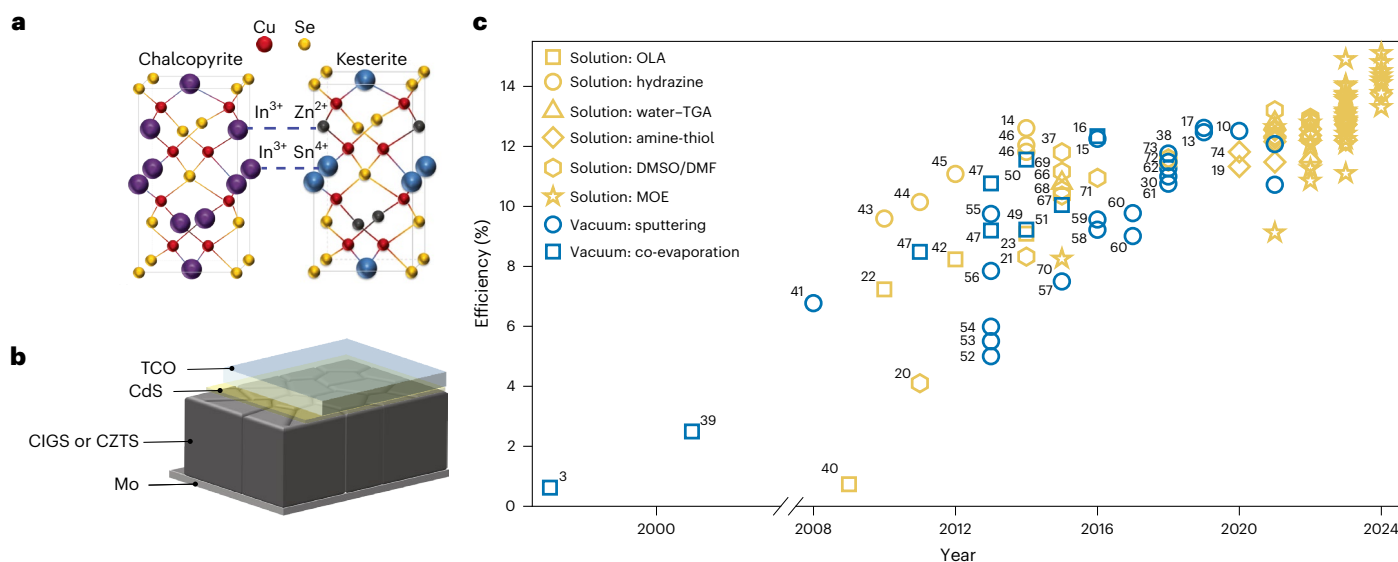


Fig. 1 | State of the art of kesterite solar cells. a, Crystal structure of chalcopyrite CIGS and kesterite CZTSSe and device structure of CIGS and CZTSSe absorber materials. **b**, Schematic of a typical device configuration for CIGS and CZTSSe solar cells. **c**, Evolution of the PCE of kesterite devices, from the first reported device to the current highest-performing one, using different synthesis strategies. The numbers represent reference citations^{3,10,13–120}. Due to space

limitations, the data points for 2021 (MOE^{29,75}, amine-thiol^{176–80}, sputtering^{64,65}, DMSO/DMF^{24,32,81,82,129} and water-TGA^{28,94}), 2022 (MOE^{84–87}, amine-thiol^{88–90} and DMSO/DMF^{25,32,82,91–93}), 2023 (MOE^{26,31,94–108,141}, DMSO/DMF^{114–116}, water-thioglycolic acid (TGA)³⁶ and amine-thiol^{109–113}) and 2024 (MOE^{18,27,33–35,117–119} and amine-thiol¹²⁰) are not labelled in the figure. TCO, transparent conducting oxide.

hydrazine-based wet processing, were adapted from Cu(In,Ga)(S,Se)₂ (CIGS) technology; however, these approaches proved inadequate in yielding high-efficiency devices, causing device efficiencies to plateau below 13% for nearly 25 years.

This Review shows that overcoming performance limitations in kesterite solar cells can be achieved through precise control and engineering of the kesterite formation pathway. Such control reduces intermediate-phase residues and prevents kinetically trapped defects (that is, residual defects present due to uncompleted chemical reactions^{1,2}). First, we comprehensively examine the interplay between precursor selection, processing conditions and device performance, establishing a chemistry–defect–performance connection. We then delve into recent advances in solution-based methods, particularly molecular ink strategies. We show that controlling molecular precursor chemistry, especially through oxidation state regulation and tailored complexation chemistry, enables the direct and uniform formation of high-quality kesterite absorber layers. Next, we show the role of pre- and post-deposition treatments, selective doping and alloying, and soft heterojunction strategies to enhance both bulk and interface quality. By drawing insights from recent high-efficiency kesterite devices exceeding 15% PCE, we outline targeted strategies to further optimize precursor chemistry, reaction pathways and defect management, aiming to drive continued improvements in device efficiency, homogeneity and reproducibility. This Review establishes a framework for precise reaction pathway engineering, enabling the design of targeted and simplified formation routes for complex multinary semiconductors. By linking precursor chemistry, intermediate phases and final microstructure, the framework provides chemical insight into phase control and opens a pathway towards more reproducible and scalable fabrication strategies. Although centred on kesterite solar cells, we believe that the concepts are broadly transferable to other emerging semiconductors, offering a platform for guided material discovery and device optimization.

Challenges and achievements in the synthesis of kesterites

Kesterite materials have a tetragonal structure that conceptually derives from the Cu(In,Ga)Se₂ chalcopyrite (CIGS) structure by substituting the

In and Ga atoms with Zn and Sn (Fig. 1a). The similar crystal structure and small changes in optoelectronic properties, such as the tunable bandgap (E_g of 1.0–2.1 eV) by alloying with isovalent elements (Ag, Cd, Ge or S) and high absorption coefficient ($\alpha \sim 10^4 \text{ cm}^{-1}$), indicate a high potential of the kesterite structure for photovoltaic applications. The typical [Mo/Absorber/CdS/(i-ZnO)/In₂O₃/SnO₂, 90:10-ITO/Ag or Ni+Al/(MgF₂), (where different layers are separated by dashes and parentheses indicate optional layers)] device structure developed for CIGS was adopted (Fig. 1b).

The first kesterite solar cell device reported by Katagiri et al.³, fabricated using PVD techniques in 1997, showed an efficiency of 0.62%. The substantial performance gap between kesterite and chalcopyrite solar cells is fundamentally linked to substituting (Ga/In)³⁺ with Earth-abundant Zn²⁺ and Sn⁴⁺ (refs. 4,5). This substitution modifies the chemistry of the absorber, introducing complex chemical landscape and reaction pathways from multiple cationic oxidation states, which creates conditions that favour the formation of defects, leading to recombination losses that limit photovoltaics performance. The initial synthesis approaches for kesterite materials were inspired by CIGS technology, relying on PVD techniques or hydrazine-based wet methods. However, the added chemical complexity introduced by the cationic substitution has led to dominant defect-related losses, resulting in a prolonged period of efficiency stagnation below 12.6% PCE.

Defects in kesterite and their impact on device performance

Defects can be classified according to their dimensionality (from three dimensional (3D) to 0D), each having a different impact on device performance, as discussed in Box 1. Throughout the initial years of investigating kesterite technology, the research focused on developing strategies to identify, mitigate and suppress the detrimental defects that critically impact device performance^{4,6,7}. Synthesis methods to achieve high-quality kesterite materials have been extensively explored through continuous trial and error. The multinary nature of kesterite materials, with multivalent Cu and Sn elements, often leads to by-product defects from incomplete or uncontrolled solid-state reactions^{1,2}. The synthesis challenges are further compounded by the involvement of volatile Sn–X (X = S or Se) and highly stable ZnX

BOX 1

Defects in kesterite materials and their impact on device performance

Defects are structural features that disrupt the discrete long-range translational periodicity of a crystal and span all dimensionalities, each with distinct implications for device performance, as presented in the inset table. 3D defects include residual secondary phases, often micrometre sized, with compositions such as Cu–Se, Sn–Se or Cu–Sn–Se. These narrow-gap, highly conductive phases are typically suppressed using Cu-poor, Zn-rich stoichiometry. Wide-gap phases such as ZnSe have been detected in devices exceeding 8% PCE¹⁶³. 2D defects, including interfaces with secondary phases, transport layers and grain boundaries are also relevant. Strategies such as chemical etching, thermal treatments and interface passivation layers aim to reduce recombination at these interfaces^{25,147–151,163}. The chemistry and role of grain boundaries remain only partially understood; different grain boundary types can coexist within the same film and their composition and orientation are independent of bulk stoichiometry^{34,37,164–167}. Stacking faults form in the {112} direction

to compensate for small deviations from stoichiometry and act as electron barriers^{11,168}. 1D defects such as dislocations and twin boundaries have been observed, but their impact remains unclear. OD defects and defect clusters are abundant due to the multinary nature of the CZTSSe system. Donor Sn-related defects (Sn_{Zn} and Sn_{Cu}) limit the intragrain minority carrier lifetime^{169,170}. Cu-related defects (V_{Cu} , Cu_{Zn} and Zn_{Cu}) control the majority carrier density^{8,171,172}. Clusters, such as $[\text{Cu}_{\text{Zn}} + \text{Zn}_{\text{Cu}}]$ are related to structural disorder: in-plane $[\text{Cu}_{\text{Zn}} + \text{Zn}_{\text{Cu}}]_{\parallel}$ and out-of-plane $[\text{Cu}_{\text{Zn}} + \text{Zn}_{\text{Cu}}]_{\perp}$ are linked to structural disorder detected by neutron diffraction and to bandgap fluctuations from the kesterite-to-stannite transition, respectively. The latter has a more pronounced impact on device performance. It has been shown that Cu–Zn disorder reduces the defect formation energy¹⁷⁰. $[\text{Sn}_{\text{Zn}} + 2\text{Cu}_{\text{Zn}}]$ clusters cause Urbach tailing due to the high impact on the band position. Fluctuations in the composition, defects and polytype can hinder charge carrier transport and enhance recombination¹⁷³.

Defects in kesterite

Dimension	Origin	Effect	Link with formation pathway	Mitigation strategies
3D	Cu–Se, Sn–Se and Cu–Sn–Se phases	Recombination Device shunting	Residue from liquid-assisted grain growth	Cu-poor and Zn-rich composition Formation pathway control
	Zn–(S,Se)	Active area loss Series resistance	High stability of ZnSe High Zn–Se affinity	Chemical etching Formation pathway control
	Cu–Zn–Sn–Se polymorphism	Recombination	Cu–Zn isoelectronic	Isovalent alloying Formation pathway control
2D	Kesterite/CdS interface	Recombination	Surface termination and stoichiometry	After treatment Passivation layer Isovalent surface alloying Formation pathway control
	Kesterite/MoSe ₂ interface	Recombination	Kesterite decomposition Morphology	Passivation layers Formation pathway control
	Grain boundaries, stacking faults and twin boundaries	Recombination	Density of grain boundary Properties of grain boundary	Extrinsic doping After treatment Formation pathway control
1D	Dislocations	Electron barriers	Unknown	Unknown
OD	Sn-related point defects (Sn_{Zn} and Sn_{Cu})	Recombination	Composition fluctuation Sn–Se volatility Growth under Zn-poor and p-type conditions	Sn or Sn–Se provided during growth Formation pathway control
	Cu-related point defects (V_{Cu} , Cu_{Zn} and Zn_{Cu})	Hole density Recombination	Composition fluctuation	Extrinsic doping and alloying Formation pathway control
	Defect clusters	Band and Urbach tailing Recombination	Composition fluctuation	Extrinsic alloying Formation pathway control

intermediate phases⁸. Detecting and quantifying many of the defects is difficult due to the complexity of the CZTSSe system, and interpreting their effects on device performance is not straightforward, as discussed in Box 1 (ref. 9). Moreover, CZTSSe devices with similar phase, composition and morphology can exhibit very different PCEs due to the impact of detrimental residual defects, which are closely related to the material formation pathway¹⁰. To comprehend how the reaction pathway can lead to grains with non-uniform composition and different concentrations of kinetically trapped defects, the thermodynamics and kinetics of the re-crystallization process should be deeply understood^{1,2,10–12}.

Reaction pathway engineering and defect formation
From Box 2, it can be observed that despite the high reactivity of the metallic precursors and thermodynamic stability of the kesterite phase, the participation of intermediate phases during the formation pathway of kesterite is unavoidable. Solid-state reactions occur only at the interfaces between two intermediate phases, which can be limited by their differing reactivities and diffusivities^{1,2}. Additionally, the formation of intermediate phases that compete with the kesterite phase may consume system energy, leaving insufficient driving force to complete the reaction. Therefore, the multiple

BOX 2

Reaction pathway engineering for understanding efficiency limits

Hydrazine pathway

Hydrazine dissolves metal, metal chalcogenides, sulfur and selenium. Hsu et al.¹⁷⁴ demonstrated the formation of $\text{Cu}_2\text{Sn}(\text{S,Se})_x(\text{N}_2\text{H}_4)_z$ frameworks ($n=4$ or 5), with Cu^+ and Sn^{4+} as the main oxidation states, and $\text{Zn}(\text{S,Se})(\text{N}_2\text{H}_4)$ dispersible particles forming a slurry^{174–176}. These frameworks mimic the oxidation state and tetrahedral geometry of the ions in kesterite, enabling rapid, strain-free transition to $\text{Cu}_2\text{Sn}(\text{S,Se})_3$ and ZnSe , with fast grain growth promoting compositional uniformity and phase homogeneity. The stability of $\text{Zn}(\text{S,Se})$ and low solid-state diffusivity of Zn limit the rate of the reaction shown in equation (1). Introducing a soluble Zn salt yields a homogeneous solution, allowing direct CZTS formation without a kinetic barrier^{14,177}.

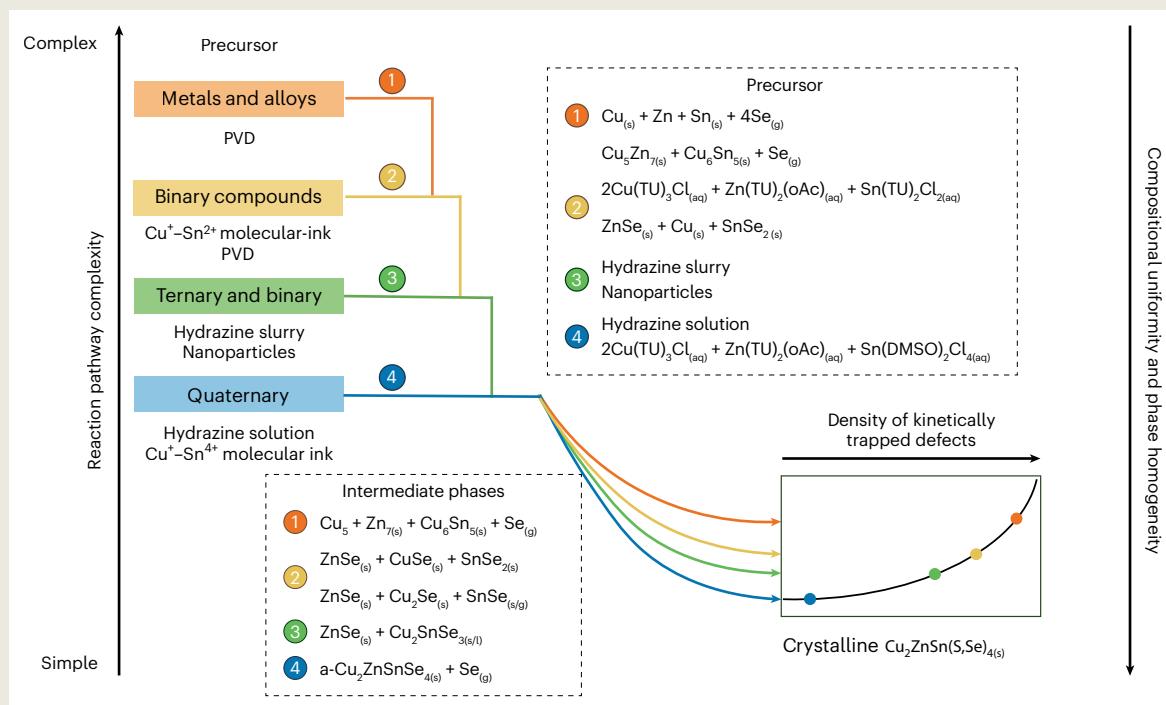
 H_2S pathway

The introduction of H_2S gas prevents Zn volatilization from the Zn/Cu/Sn stack by forming $\text{Zn}(\text{S,Se})$, whereas the Cu–Sn alloy remains unreacted beneath the $\text{Zn}(\text{S,Se})$ layer. Kesterite forms via the reaction of Cu_2SnSe_3 (CTSe) and $\text{Zn}(\text{S,Se})$ ^{13,17}. The low-temperature formation of the CTSe phase prevents volatile Sn–Se phase losses, without the use of additional Sn or $\text{Sn}(\text{S,Se})_x$

powders, which result in surface Sn–Se secondary-phase residues^{178,179}. Liquid CTSe promotes grain growth through liquid-phase transport, enabling large grains and high compositional homogeneity¹⁸⁰. However, CZTSSe grain formation remains kinetically limited by the reaction shown in equation (1), and residual liquid-CTSe can degrade the performance due to its low E_g and high carrier concentration.

Soft-selenization pathway

The soft-selenization pathway employs a two-zone thermal chamber with separate temperatures for Se gas and samples, as well as a Mo cracking barrel to control the Se allotrope^{10,181}. Zn/Cu/Sn/Cu/Zn metallic stacks form Cu–Zn and Cu–Sn alloys, which decompose at 280°C under reactive Se, forming Cu_2SnSe_3 and ZnSe . This enables kesterite formation at very low temperature (350°C) under Sn^{4+} -rich conditions. Ge doping further stabilizes the complete decomposition of Cu–Zn alloys, favouring the CZTSe phase through the reaction shown in equation (1)^{38,182}. Na is critical for grain growth under low-Se-vapour pressures, whereas Zn up-diffusion forms a ZnSe layer that passivates the front interface^{10,125,152}.



Formation pathway of kesterite materials. Schematic of the different reaction pathways to achieve the formation of kesterite absorbers. The total free energy is represented qualitatively on the y-axis. The x-axis qualitatively represents the lattice configurations of the molecules and solids, which are different for all of the phases represented in the figure. The amorphous kesterite intermediate phase is labelled as $\text{a-Cu}_2\text{ZnSnSe}_4$. As shown in the picture, synthesis pathways through the intermediate phases can lead to kinetically trapped defects due to Sn loss, as well as non-uniform nucleation, reaction and growth rates. Therefore, careful control of the synthesis conditions is required to achieve high-quality, compositionally uniform and homogeneous materials. In contrast, direct formation of the kesterite phase facilitates achieving high-quality, compositionally uniform and homogeneous materials.

intermediate phases contribute to compositional non-uniformity and the formation of kinetically trapped defects. The compositional complexity of kesterite compounds is related to an intricate reaction pathway and the participation of highly stable (ZnX) and volatile

(Sn–X) intermediate phases, which can lead to excess concentration and fluctuations of kinetically trapped defects^{1,2,12}. This is a common occurrence in the synthesis pathway of kesterite films from PVD-based methods.

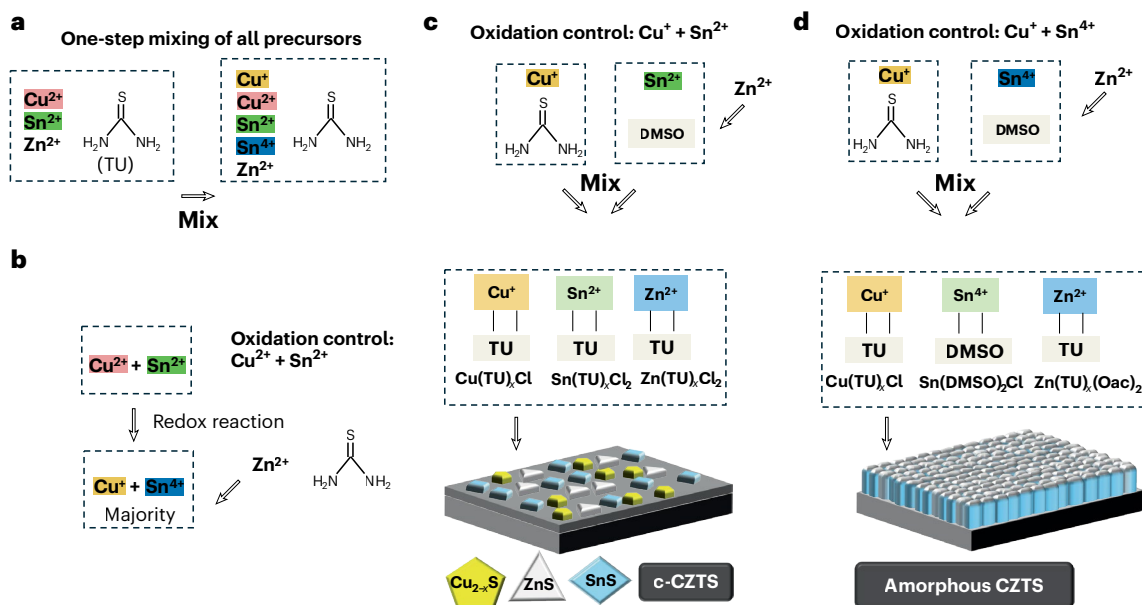
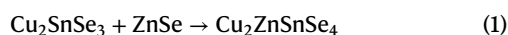


Fig. 2 | Solution-controlled oxidation state, from molecular precursors to precursor film. **a**, Schematic of oxidation state evolution when metal salts are mixed in DMSO simultaneously. Dissolving all of the salt precursors simultaneously leads to a translucent yellow colour, indicating the presence of Cu^{2+} and S species. The uncontrolled oxidation state and molecular geometry of Cu and Sn lead to an inhomogeneous precursor film²¹. **b**, Schematic of oxidation state evolution through a controlled redox reaction of Cu^{2+} and Sn^{2+} by dissolution of salt precursors. The oxidation state of Cu is reduced completely to Cu^+ by Sn^{2+} . The excess Sn^{2+} might result in the presence of residual solid-state intermediate phases. A translucent and light yellow solution results from this process²¹. **c**, Schematic of the molecular complexes formed using Cu^+ and Sn^{2+} precursors. Due to their soft nature, Cu^+ and

Sn^{2+} form complexes with thiourea, leading to an undesired oxidation state and geometry of Sn. The uncontrolled oxidation state and molecular geometry of Cu and Sn lead to an inhomogeneous solid-state precursor film, similarly to those described by Ki and Hillhouse²⁰ and Gong et al.¹²⁹. **d**, Schematic of the molecular complexes formed using Cu^+ and Sn^{4+} precursors. The disproportionation of the Cu^+ oxidation state in DMSO is avoided by the introduction of thiourea. The reduction of Sn^{4+} by thiourea is avoided by the dissolution of SnCl_4 directly in DMSO, forming highly stable complexes that preserve the oxidation state with the desired geometry. A translucent and light yellow solution results from this process¹²⁹. The controlled oxidation state and molecular geometry of Cu and Sn lead to a homogeneous solid-state precursor film. TU, thiourea.

To highlight the importance of precise control over the formation pathway for achieving homogeneous and compositionally uniform kesterite absorbers, leading to high-efficiency devices, the formation pathways of the highest-efficiency solar cells currently reported are presented and discussed in Box 2. The complexity inherent to multinary compounds necessitates optimizing material synthesis based on the precursor chemistry and exploring different experimental conditions to identify and understand each formation pathway and its kinetic limitations. The understanding of the CZTSSe system led to three successful pathways achieving efficiencies beyond 12.5% and six achieving efficiencies beyond 12% before the synthesis method discussed in this Review^{10,13–17}.

As shown in Fig. 1c, Todorov, Mitzi and colleagues¹⁴ set a long-lasting record of 12.6% in 2013 using a wet process method based on hydrazine solvents and a fast thermal treatment, with samples introduced in a home-made oven at 540 °C. Later, five different PVD methods achieved efficiencies beyond 12%, with two of them surpassing 12.5% by processing the materials under very mild conditions with low temperatures and chalcogen partial pressure^{10,17}. In Box 2, successful PVD and solution-based synthesis processes are described and compared. The different processing conditions determine the reaction pathway that reduces the concentration of detrimental defects, improving the optoelectronic properties of the CZTSSe absorber as well as the related device performance.



In summary, for highly reactive precursors, such as metals (PVD routes), kesterite formation through the reaction shown in equation (1) should be achieved at relatively low temperatures (below 400 °C)

and low Se partial pressure to suppress the nucleation and grain growth of intermediate phases without hindering kesterite phase formation (Box 2). This step must be followed by a relatively low-temperature and high-Se-partial-pressure stage to favour grain growth and Ostwald ripening while preventing kesterite decomposition. This approach avoids detrimental effects of the participation of intermediate phases such as Sn loss, compositional non-uniformity, phase non-homogeneity and therefore kinetically trapped defects. In contrast, homogeneous and reactive molecular precursors from hydrazine are directly introduced under high temperature and chalcogen partial pressure, resulting in fast and barrierless kesterite formation with grain growth and Ostwald ripening stages, yielding highly homogeneous kesterite films with a low density of kinetically trapped defects. The contrast between the processing conditions required to achieve high-quality kesterite films as a function of the precursor chemistry highlights the chemistry–defect–performance relationship of multinary compounds.

The molecular ink pathway

Role of molecular ink design

Controlling the complex reaction pathway is critical for achieving high-efficiency devices by precisely managing the compositional uniformity and phase homogeneity of the CZTSSe layer at the micro- and nanoscale. In parallel with hydrazine- and PVD-based processes, solution methods utilizing comparatively safer and more environmentally friendly solvents, such as dimethylsulfoxide ($(\text{CH}_3)_2\text{S}-\text{O}$ (DMSO)), *N,N*-dimethylformamide ($(\text{CH}_3)_2\text{N}-\text{CHO}$ (DMF)), 2-methoxyethanol ($\text{CH}_3-\text{O}-(\text{CH}_2)_2\text{OH}$ (MOE)) and various amine-thiol systems, have been developed, offering reduced toxicity and improved handling for scalable applications. As shown in Fig. 1c, the use of organic solvents has progressively improved the performance since 2011 and has now

Table 1 | Molecular complexes formed in DMSO–thiourea solution for different kesterite metals with different oxidation states

Cation (Lewis acid)	Electronic configuration	Ionic radius (Å)	Hard or soft acid	Preferred ligands	Cation geometry	Reference
Cu ⁺	[Ar] 3d ¹⁰ 4s ⁰	0.46–0.77	Soft	Soft (S) Covalent	Tetrahedral Cu(TU) ₃ Cl Homodirected	Ref. 128
Cu ²⁺	[Ar] 3d ⁹ 4s ⁰	0.57–0.73	Borderline hard	Hard (O) Ionic	Square pyramidal CuCl ₂ (DMSO) ₂ Hemidirected	Refs. 131,132
Sn ²⁺	[Kr] 4d ¹⁰ 5s ²	1.18	Soft	Soft (S) Covalent	See-saw Sn ₂ (TU) ₅ Cl ₄ Sn(TU)Cl ₂ Hemidirected	Refs. 133–135
Sn ⁴⁺	[Kr] 4d ¹⁰ 5s ⁰ 5p ⁰	0.55–0.81	Borderline hard	Hard (O) Ionic	Octahedral Sn(DMSO) ₂ Cl ₄ Homodirected	Ref. 128
Zn ²⁺	[Ar] 3d ¹⁰ 4s ⁰	0.60–0.90	Borderline soft	Soft (S) Covalent	Tetrahedral Zn(TU) ₂ (CH ₃ COO) ₂ Homodirected	Refs. 128,132,162
Ag ⁺	[Kr] 4d ¹⁰ 4s ⁰	1.00–1.28	Soft	Soft (S) Covalent	Tetrahedral Ag(TU) ₃ Cl Homodirected	Ref. 132
Li ⁺	[He] 2s ⁰	0.59–0.92	Hard	Hard (O) Ionic	Unknown	
Cd ²⁺	[Ar] 4d ¹⁰ 5s ⁰	0.78–1.10	Soft	Soft (S) Covalent	Tetrahedral Cd(TU) ₂ Cl ₂ Homodirected	Ref. 162
Ge ⁴⁺	[Ar] 3d ¹⁰ 4s ⁰	0.39–0.53	Hard	Hard (O) Ionic	Unknown	

TU, thiourea.

enabled efficiencies of 15% (0.27 cm²) and 13.5% (>1 cm²), raising the interest of the photovoltaics community towards this reproducible and scalable synthesis method^{3,10,13–120}.

We use (Lewis) hard and soft acid and base theory to understand the stability of molecular compounds and the subsequent reaction mechanism and formation pathway leading to high-quality CTZSSe. An acid is a chemical species that accepts electrons, whereas a base donates electrons, resulting in the formation of bonds from acid–base reactions. The hardness of the chemical species relates to the polarizability of the molecule, with hard species being non-polarizable and soft species being strongly polarizable. Bonds between hard acid–base species are stable and ionic, whereas bonds between soft acid–base species are stable and covalent. Hard–soft combinations are typically unstable. Steric and redox reaction effects must also be considered for a complete picture of the solution-based precursors.

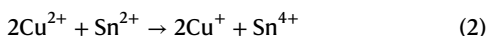
Similarly to the N sites in hydrazine, these solvents have an O site that can act as a Lewis base. Despite the relative hardness of the O Lewis base sites of the solvents, the complexation and solubility of metal, metal oxides and chalcogenide precursors is extremely challenging and can only be effectively done with amine–thiol solvents¹⁹.

Taking this into consideration, in 2011, Ki and Hillhouse²⁰ utilized a chemical route by spin coating a DMSO-based solution of highly soluble, inexpensive and commercially available metal salts and thiourea (SC(NH₂)₂), where thiourea was used as a source of S upon thermal decomposition. The DMSO–thiourea system possesses a wide variety of Lewis base sites with different hardnesses (that is, the O site in DMSO and N sites in thiourea are hard, whereas the S sites in thiourea and DMSO are soft, with the latter being typically sterically hindered)¹²¹. Therefore, the formation of soluble Lewis's acid–base molecules should be expected, which upon thermal decomposition, can be used as molecular precursors for the synthesis of CZTS thin films. As described in Fig. 2a, the process of Ki and Hillhouse²⁰ yielded efficiencies above 4%, but the performance was still far from that of the hydrazine-based process.

The initial oxidation state of Cu and Sn differs from the oxidation state in the kesterite phase, leading to molecular precursors with non-tetrahedral geometries, as described in Table 1. In the tetragonal kesterite structure, the first-neighbour coordination of the ions is tetrahedral, whereas in common Cu²⁺- and Sn²⁺-based solid-state secondary phases, the coordination is typically non-tetrahedral. In 2014, Xin et al.²¹ provided experimental evidence of the correlation between improved solar cell performance and the control of oxidation states by thoroughly understanding the redox reactions, controlled by the mixing order. This finding underscores the importance of preserving the local structural environment of the constituent elements. Specifically, it suggests that the synthesis pathway that minimizes local coordination rearrangements from the molecular precursor to the crystalline absorber facilitates the direct formation of kesterite or other tetrahedrally coordinated structures. Conversely, precursors with mismatched geometries may lead to kinetically trapped defects involving ions that fail to adopt the necessary tetrahedral coordination, limiting material quality and device performance.

In the work of Xin et al.²¹, as schematized in Fig. 2b, Cu(CH₃COO)₂·H₂O was first added in DMSO. Then, SnCl₂·2H₂O was added to the vial, where the redox reaction shown in equation (2) was proposed to take place. This reaction is potentially shifted towards the product by the formation of Sn⁴⁺ complexes with the solvent. The oxidation reaction of Sn²⁺ has been detected for precursor species in Sn-based perovskite molecular inks using DMSO and DMF as solvents^{122,123}. Additionally, due to the absence of thiourea in the solution, the well-known reduction of Sn⁴⁺ to oxidize thiourea into formamidine disulfide (NH₂(NH)CSSC(NH)NH₂), which decomposes to thiourea, cyanamide (CNNH₂) and elemental sulfur, is suppressed. It has been shown that Cu²⁺ can lead to the same thiourea oxidation mechanism¹²⁴. Thus, after one night of stirring, the solution colour changed from deep blue to translucent light yellow, indicating a reduced concentration of Cu²⁺. The persistence of the Cu²⁺ oxidation states relates to the disproportionation mechanism of Cu⁺ to form Cu⁰ and Cu²⁺ in polar solvents, as demonstrated by

Clark et al.¹²¹, unless an additive that reverses the reaction is included. Interestingly, the addition of ZnCl_2 and thiourea further stabilized the Cu^+ oxidation state, eventually removing all Cu^{2+} , as indicated by a light yellow solution¹²¹. However, as shown in Fig. 2b, the typical Cu-poor stoichiometry leads to residual Sn^{2+} content in the precursor molecular ink, which might result in the participation of intermediate phases during kesterite growth.



In contrast, when the precursor salts are mixed at the same time, as represented in Fig. 2a, the presence of Cu^{2+} and S species in the solution leads to a yellow colour. This suggests that Cu^{2+} (and therefore also Sn^{2+}) does not completely undergo the self-regulating redox reaction, oxidizing the thiourea molecules, resulting in an uncontrolled molecular precursor oxidation state and molecular geometry, as well as an inhomogeneous and defective absorber. In the work of Xin et al.²¹, the clear molecular precursor solution was spin-coated and annealed at 540 °C on a hotplate, resulting in a partially crystalline, yet not amorphous, precursor layer, compared with the inhomogeneous precursor film described by Ki and Hillhouse²⁰. This precursor film was then selenized using a vented graphite box at 540 °C under an atmosphere of Se. The reduced surface-to-volume ratio leads to an enhanced stability of the crystalline CZTS (c-CZTS) precursor layer, which requires a higher activation energy. This can result in a more inhomogeneous and non-uniform nucleation stage, hence leading to inhomogeneous and non-uniform phase and element distributions. No interface passivation strategy was used in this work, leading to a reported device performance of below 9%.

Similarly to the partially crystalline CZTS system, the family of nanocrystalline CZTS (nc-CZTS) precursors, prepared via the hot injection method using oleylamine as the solvent, shows how thin-film precursor crystallinity and carbon content affect performance. The reduced reactivity of the nc-CZTS precursors, combined with the larger amount of organic matter, leads to a higher energy barrier for nucleation and grain growth, leading to potential inhomogeneities at the micro- and nanoscale. Additionally, decomposition of the nc-CZTS precursor into the intermediate species was observed and proposed to drive the grain growth in this system¹²⁵. Therefore, even though the phase transformation of the nc-CZTS system appears to be kesterite to kesterite, it involves the participation of intermediate phases (that is, the reaction shown in equation (1)), leading to a maximum PCE of 9%^{22,23,126,127}. Interestingly, in this system it was observed that working under very highly saturated Se atmospheres to achieve grain growth was necessary, unless assisted by alkali metals¹²⁵. Compared with hydrazine processes, the methods developed using organic solvents seem to be kinetically limited by the reaction pathway. Organic molecules must be decomposed into volatile species to be removed. In contrast, the hydrazine solvent, which decomposes at very low temperatures and very exothermically, allowed for the fast decomposition of small molecules, nucleation and grain growth without kinetically limiting processes. Consequently, it seems critical to preserve compositional uniformity, homogeneity and the oxidation state of the molecular inks, and for this, the re-crystallization step should be rapid. To achieve this, a solvent-drying step at low temperature and a subsequent selenization of the precursor film needs to be implemented. However, it is evident that the formation of crystalline CZTS precursors can introduce a high energy barrier for the nucleation, leading to decomposition of the precursor film and non-homogeneous nucleation, resulting in devices far from record-level performance.

Gong et al.^{24,25,128,129} achieved efficiencies beyond 12.6% by deeply understanding the role of molecular precursors in their synthesis process, as shown in Fig. 2c,d. However, several critical factors must be discussed to understand the roles of the molecular ink and the mixing order, which determine the formation of different acid–base

complexes in solution, leading to highly homogeneous and reactive molecular precursors. Preserving the uniformity and homogeneity of the precursor solution after the re-crystallization process is essential. In the work developed by Gong et al.^{24,25,128,129}, two separate solutions were prepared and then mixed later, as schematized in Fig. 2c,d. In the first vial, thiourea was dissolved in DMSO, then CuCl was added. In the second vial, first SnCl_4 was dissolved in DMSO, followed by the addition of $\text{Zn}(\text{CH}_3\text{COO})_2$. Finally, the second vial was poured into the first to obtain the precursor solution. This process preserves the oxidation state of the precursor salts in the ink molecular species. The mixing order is critical to preserve the desired Cu^+ and Sn^{4+} oxidation states for the formation of a single-phase CZTS layer. As proposed previously, it has been demonstrated that thiourea can stabilize the Cu^+ oxidation state in DMSO and DMF, increasing the solubility of Cu^+ salts in the solvent^{121,130}. Without using thiourea as an additive, the affinity of the hard Cu^{2+} ions with the hard O sites of DMSO can lead to the formation of Cu^{2+} –DMSO complexes^{131,132}. Instead, with thiourea, the soft base S sites tend to preferentially complex with soft Cu^+ acid species, leading to CuCl –thiourea complexes, as shown in Table 1 (ref. 128).

It is also widely known that thiourea can complex with Zn^{2+} and Sn^{2+} salts, as shown in Fig. 2c refs. 133–136. Adding the Zn precursor salt to vial 1 turns the solution dark blue, which indicates the presence of Cu^{2+} in the solution. This suggests a reduction in the complexation of Cu with thiourea, leading to the disproportionation reaction. This is possibly due to the competition of Cu and Zn to bond with thiourea, as well as the addition of hard O sites from the acetate molecule resulting in the formation of stable Cu^{2+} – $(\text{CH}_3\text{COO})_2$ /DMSO complexes. Therefore, adding the Zn precursor into vial 2 is necessary to preserve the Cu^+ oxidation state. Furthermore, using $\text{Zn}(\text{CH}_3\text{COO})_2$, which is more soluble in DMSO compared with ZnCl_2 , enables higher solubility of the Sn –DMSO complex. Similarly to the Cl^- exchange stabilizing Cu^+ , exchange of $(\text{CH}_3\text{COO})_2^-$ and Cl^- between SnCl_4 and $\text{Zn}(\text{CH}_3\text{COO})_2$ is expected due to the stronger hardness of Sn compared with that of Zn, leading to a preferential bonding of Sn with O instead of Cl, resulting in a stable solution when the vials are mixed.

Sn^{2+} –thiourea complexes are seemingly more stable than Sn^{2+} –DMSO complexes, which is related to the softness of Sn^{2+} , leading to preferential bonding with soft S base sites, as shown in Table 1. The formation of Sn^{2+} –DMSO coordination complexes through the formation of Sn–S bonds instead of Sn–O bonds has been detected¹³⁰. Complexes from the Lewis acid–base reaction of SnCl_4 with thiourea have been detected; however, the molecular complexes of Sn^{4+} are not stable due to the previously mentioned thiourea oxidation leading to Sn^{2+} (ref. 137). By first mixing SnCl_4 and DMSO, highly stable $\text{Sn}(\text{DMSO})_2\text{Cl}_4$ complexes are formed¹²⁸. The high stability of the Sn^{4+} –DMSO complex is expected due to the formation of a hard–hard acid–base complex and is confirmed by the very exothermic reaction, as indicated by the heat released when DMSO and SnCl_4 are mixed. The formation of this complex prevents oxidation of the thiourea to formamidine disulfide by a reduction of Sn upon mixing with thiourea; therefore Sn–thiourea complexes are not detected. Thus, the formation of stable complexes with the proper oxidation state has been demonstrated to be critical for the achievement of high-efficiency solar cells. This probably relates to the geometry and thermal behaviour of the molecular precursors.

Nevertheless, the evolution process from the molecular precursors to the crystalline absorber layer is not yet fully understood. Metal halide precursors, such as ZnCl_2 and SnCl_4 , would not be suitable for CZTS thin-film deposition due to their high volatility, and the formation of molecular complexes with low vapour pressure can prevent elemental losses¹²¹. However, would any soluble molecular precursor that preserves the desired stoichiometry with low volatility be sufficient to produce the desired single kesterite phase? This seems not to be the case either. To shed light on this, understanding the coordination chemistry of ions with an ns^2 electron lone pair (such as Pb^{2+} and Sn^{2+}) is essential. The lone pair will typically lead to hemidirected complexes

(hemi indicates half), where all of the bonds are affected by the lone pair. Typically, in hemidirected complexes there is an anti-bonding metal-ion donor atom interaction in which there is a gap in the distribution of bond lengths to the ligands. This leads to complexes with low coordination numbers, characterized by a wide and gapped metal–ligand bond length probability distribution, which in the case of Sn^{2+} leads to undesired pyramidal geometries¹³⁴. Hence, as shown in Fig. 2a,c, a very non-uniform and reactive molecular precursor ink is obtained, which can lead to inhomogeneous thin films. This results in the formation of an intermediate species, also with different cation geometries, along the reaction pathway, as shown in Box 2. In contrast, the empty $5s^0$ lone pair and smaller atomic size of Sn^{4+} allow formation of the more ionic Sn–O bond and formation of the Sn^{4+} –DMSO complex with a higher coordination number (octahedral geometry), leading to holodirected (holo indicates complete), much narrower and ungapped bond length probability distribution, thus enabling direct formation of the CZTS phase, as schematized in Fig. 2b,d.

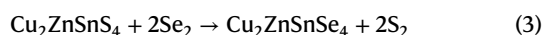
As shown in Table 1, the complexes of Cu^+ and Zn^{2+} with thiourea have tetrahedral and holodirected geometries, leading to a homogeneous and uniform bond distribution, which can enable a uniform CZTS precursor. The formation of metal–organic frameworks with a chemical environment similar to the kesterite phase has not yet been detected in this system, suggesting that a homogeneous and uniform molecular solution could be sufficient to achieve the kesterite phase uniformly. Further investigation is necessary to unveil the intricacies of the molecular precursors. Therefore, as shown in Table 1, experimental evidence indicates that precise control of Cu and Sn oxidation states is crucial for determining the geometry and decomposition dynamics of the molecular ink precursor, which subsequently determines the critical reaction pathway of Box 2.

The same complexes have been confirmed for DMF as a solvent; however, it has been suggested that with MOE a sol–gel is formed¹³⁸. Nevertheless, no precipitates are observed after the centrifugation of MOE-based solution precursors, suggesting a homogeneous precursor solution. To shed light on this, the interaction between SnCl_4 and MOE molecules should be further explored. Adding $\text{HCl}/\text{H}_2\text{O}$ (37%) prevents the alcoholysis of MOE, thus avoiding sol–gel formation, resulting in a more uniform molecular precursor ink and subsequent nucleation and grain growth, which contributes to achieving high-efficiency devices beyond 14%¹³⁹. Thin-film precursors obtained by DMSO and DMF solvents are very similar due to the similar boiling points and molecular precursor size. In contrast, MOE solvent has a lower boiling point and a larger molecular precursor size, leading to a less compact precursor. This potentially results in a more reactive MOE thin-film precursor that can nucleate more homogeneously and reduce the kinetic barrier for grain growth and Ostwald ripening stages by enabling more efficient C removal, as suggested by Fig. 1c and discussed in the following section. This reactivity leads to slightly better device performance. Additionally, the low hygroscopicity of these solvents allows for easy solution preparation and spin-coating steps under an ambient atmosphere, making them suitable for industrial production processes.

Phase evolution towards highly efficient solar cells

In the work developed by Gong et al.^{128,129}, thermal annealing in an N_2 glovebox after spin coating enabled molecular precursor decomposition, yielding a CZTS layer directly from the molecular precursors. Interestingly, using a lower temperature (320 °C) resulted in the formation of an amorphous CZTS phase, which is much more reactive than crystalline CZTS due to the higher strain and surface area. Compared with the partially crystallized precursors, the apparent barrier for the nucleation of the crystalline CZTSSe phase is lowered, enabling much more homogeneous and uniform CZTSSe nucleation, preventing decomposition into the intermediate species and resulting in a high-quality material. However, compared with the metallic and metal chalcogenide precursors which have higher free energies than either

the amorphous or crystalline kesterite phases, the reaction shown in equation (3) has a positive reaction enthalpy (61 kJ mol^{−1}) and must be an entropy-driven reaction¹⁴⁰. To shift the reaction towards the product, a high quantity of Se must be provided (>300 mg in a 36 cm³ graphite box and a 5,000 cm³ tube), as well as sufficient temperature to achieve a uniform grain growth and Ostwald ripening stage.



As shown by Gong et al.^{128,129}, using DMSO as a solvent, the polycrystalline CZTSSe nucleation mechanism starts at the top and bottom of the thin film almost simultaneously, indicating that both temperature and Se are critical. Using Sn^{2+} precursors instead of Sn^{4+} results in a precursor film containing intermediate phases, and the reaction pathway involves their participation with different interfaces, reactivities and diffusion rates, as shown in Fig. 3a refs. 128,129. This leads to the same phase, morphologies and composition as for the Sn^{4+} precursor, but results in very different device performance, due to the residual defects associated with a different reaction pathway, similar to the work of Li et al.¹⁰. Using MOE as a solvent, the final morphology of the absorber film is also a bi-layer structure, with large grains at the top and a fine grain layer at the bottom, indicating that the nucleation and grain growth are different at the top and bottom interfaces. As the free energy of the amorphous CZTS precursor is much lower than that of the metals and the hydrazine precursors, it must undergo an activation step. Understanding of the selenization stage was further improved by Zhou, Meng and co-workers^{26,27,141}. These researchers showed that the selenization process must avoid the participation of intermediate phases during grain growth, as represented in Fig. 3b. Strategies include minimizing the partial pressure of selenium (p_{Se}) during the ramping stage by increasing the total pressure of the chamber and using fast ramps; achieving an ultra-high p_{Se} before the increase in the film temperature; and multinary alloying^{26,27,141}. Nevertheless, a high p_{Se} is required to drive the phase transformation.

As schematized in Fig. 4, high p_{Se} is a necessary kinetic condition to obtain a uniform nucleation stage, as the re-crystallization dynamics in the early stages of reactive annealing seem to be limited by the rate of the reaction shown in equation (3), which has a positive reaction enthalpy²⁶. Therefore, it seems that synchronized high Se pressure and high temperature are required to obtain a uniform nucleation stage of the CZTS reaction with Se to form CZTSSe, and additional strategies are necessary for avoiding the detrimental participation of intermediate phases. It has been demonstrated that isovalent alloying with Ag, Cd or Ge can reduce the nucleation barrier²⁷. Once the reaction process is completed, the top grains keep growing in the presence of Se in the atmosphere. This process is found to be limited by the diffusion of Se towards the precursor layer or the diffusion of elements of the precursor layer, which remain unreacted between the top and bottom crystalline grains, in either solid or liquid phases, consistent with the observed round-shaped crystals at the top²⁶. In contrast, re-crystallization of the bottom layer is clearly limited by the diffusion of Se. This is related to the formation of large crystalline and compact-surface kesterite grains throughout the first stages of the reaction, which limit the diffusion of Se towards the bottom layers^{26,28}.

Another possibility is that the fine-grained layer originates from the decomposition of CZTS into intermediate phases, with the posterior reaction in the presence of low concentrations of Se. However, it has been argued that the presence of fine-grained bottom layers stems from the need to remove carbon, which also might be the origin of kinetic limitation in this system, leading to the need for two thermal annealing stages, as presented in Fig. 4 (ref. 141). Carbon may not be fully removed during the low-temperature annealing stage, and the residual carbon species interact with the thin film during the subsequent high-temperature selenization process. Such interaction of C and Se was further understood by separating the thin-film

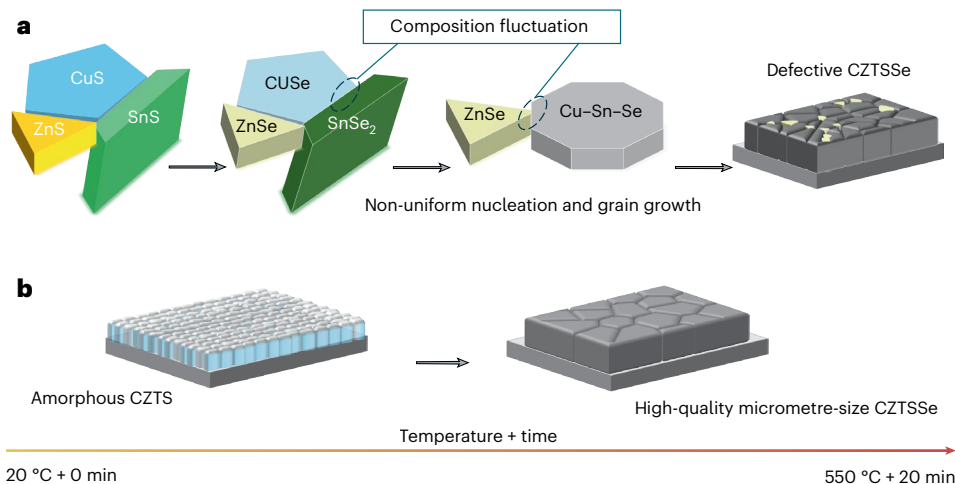


Fig. 3 | Role of precursor film in phase evolution to crystalline absorber.

a, Schematic of the non-uniformity associated with the participation of intermediate phases with different reactivities and diffusivities during the reaction pathway. The yellow features present in the defective CZTSSe

represent ZnSe secondary phases, which are typically present due to the Zn-rich composition and high ZnSe stability. **b**, Schematic of a single-phase, compositionally homogeneous CZTS precursor undergoing direct conversion into a high-quality CZTSSe absorber layer without intermediate-phase formation.

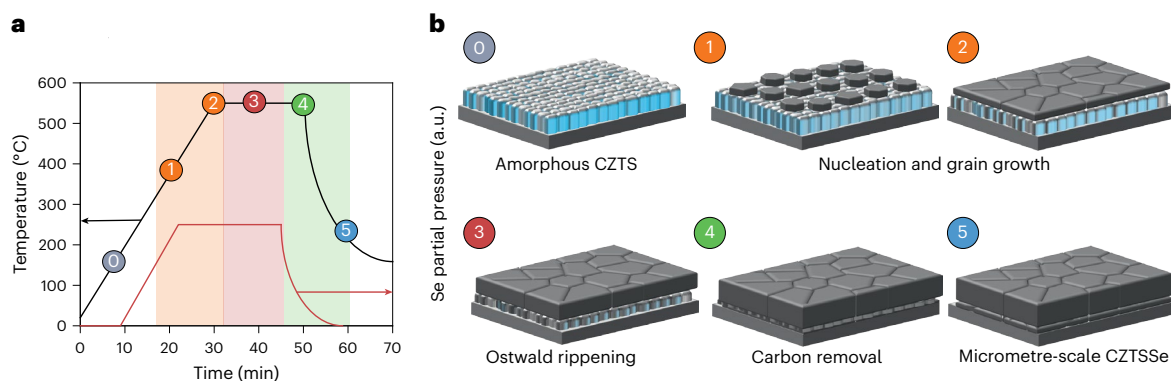


Fig. 4 | Phase evolution of amorphous film to high-quality absorber.

a, Thermal and Se vapour pressure (p_{se}) profiles during a typical thermal annealing of the amorphous CZTS precursor. **b**, Schematic of the phase transformation of amorphous CZTS to crystalline CZTSSe, as indicated in Fig. 3b. (1) Beyond 350 °C, the CZTSSe phase starts to nucleate at the front and back. (2) Grain growth of the top layer is complete. Once the top layer has completely crystallized, the kinetic limitation of the grain growth changes from reaction to diffusion. Regimens (1) and

(2) occur within the orange-shaded interval. (3) The unreacted precursor layer is partially consumed by the crystalline CZTSSe grains through the Ostwald ripening mechanism; this regimen takes place in the red-shaded area. (4) Corresponding to the green-shaded area, the p_{se} decays, allowing for carbon removal, which limits consumption of the remaining precursor layer. (5) The crystallization process is complete, resulting in the typical bi-layer structure. These concepts were extracted from refs. 26,129,141.

precursor from the Se source using a two-zone tubular furnace, revealing that the observed decay of Se vapour pressure during the latter stage of the reaction mechanism of Zhou et al.²⁶ relates to the C removal; thus, the Se vapour pressure profile should be similar to that shown in Fig. 4 (that is, high and steep at the initial stages, to suppress the precursor decomposition and start the nucleation, and lowered during the grain growth and Ostwald ripening stage, which might be limited by carbon removal, Se diffusion or precursor-layer elemental diffusion)¹⁴¹. The presence of carbon typically relates to double-layer formation; nevertheless, it has also been associated with metallic Cu and/or chalcogen deficiency in the layer^{141,142}. Aiming to unify these observations, it seems that residual carbon can lead to the formation of CSe₂, which can polymerize, forming non-volatile (CSe₂)_n species, trapping C and Se and thus reducing the local Se availability, resulting in a chalcogen deficiency and the subsequent double-layer formation¹²⁷. If the Se vapour pressure remains high throughout the reaction, carbon residues are observed, and during growth the diffusion of C, Se and the cations is limited. However, reducing p_{se} during the latter stages tends to volatilize CSe₂ species;

hence, almost no C can be observed in the layer. Interestingly, pure sulfide kesterite (c-CZTS) prepared via solution methods exhibits a single-layer morphology, most likely given that, unlike CSe₂ (see above), CS₂ cannot form polymers^{127,143}. Compared with Se compounds, as S atoms are present in a quasi-stoichiometric quantity in the precursor layer, they do not need to diffuse to drive nucleation and grain growth in c-CZTS to form highly crystalline and faceted CZTS grains^{143–146}, making the process less kinetically limited and only dependent on temperature (with a high enough S vapour pressure to avoid surface decomposition).

For device performance, the grain quality of the topmost layer is critical¹²⁸. The use of Sn⁴⁺ molecular precursors at the top layers is crucial for device performance and suggests that most of the minority carriers are generated in the top-layer grains. The bottom layer does not participate in minority carrier generation and contains a low concentration of minority carriers if the top layer is thick enough (larger than 1.0–1.2 μm), only contributing to hole transport; thus, device performance is less dependent on grain quality in the bottom layer, which was tuned using an Sn²⁺ or Sn⁴⁺ precursor.

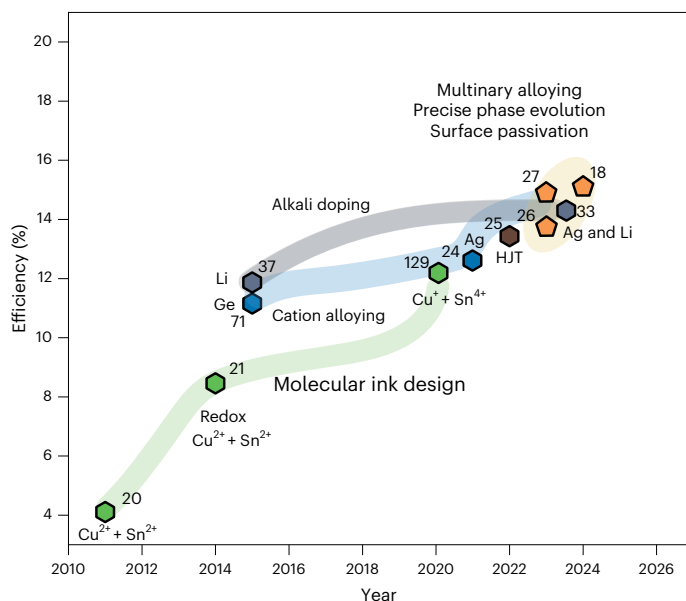


Fig. 5 | Technological milestones in kesterite solar cell development via molecular inks. This figure highlights the critical role of molecular ink design (green^{20,21,129}) in controlling oxidation states, phase evolution and PCE. Stepwise implementation and optimization of previously successful strategies, such as cation alloying (blue^{18,24,26,27,71}), alkali doping (grey^{33,37}) and heterojunction engineering (brown^{18,25–27,33}), has enabled efficiencies beyond 15%, as shown in orange. The numbers represent reference citations.

Role of soft post-deposition heat treatment

To achieve high efficiencies beyond 13%, high-quality grains are not sufficient and the application of a heterojunction heat treatment (HJT) is critical^{24,25,128,129}. There have been various successful HJT strategies in the kesterite system. The improvements observed have typically been ascribed to improvements of the CZTSSe/CdS interface, the tuning of optoelectronic properties, such as doping density, related to Cu/Zn disorder of the kesterite and/or re-crystallization of the CdS^{25,29,30,147–150}. Gong et al.²⁵ noted that, due to its chemical similarity with Cd, Zn is very soluble in NH_4OH through the formation of $\text{Zn}[\text{NH}_4\text{OH}]_2$ complexes, leading to Zn loss during chemical-bath-deposited CdS and substantial performance losses at the interface. It had been previously noted by Neuschitzer et al.¹⁴⁸ that the CZTSSe surface shifts to Zn-poor composition during chemical bath deposition and is recovered after heat treatment. High-efficiency devices fabricated at the Daegu Gyeongbuk Institute of Science and Technology, University of New South Wales and International Business Machines Corporation did not report any interface heat treatment, as well as a deposition of the window layer at room temperature. Possibly, the surface termination depends highly on the thermal treatment and gas atmosphere, suggesting that lower temperatures lead to a more compact, less reactive or more Zn-rich surface of CZTSSe. The use of ZnSe or other passivation layers, or the application of HJT, enables non-limiting interface losses and the highest-performing devices^{151,152}. Further optimization in the front interface engineering of kesterite devices seems critical to further push the PCE of this technology.

Strategies for efficiencies beyond 13%

As presented in Fig. 5, the application of strategies that had previously been successful in improving the optoelectronic properties of kesterite absorbers synthesized by the different methods presented above has led to high-efficiency kesterite devices through its implementation in the molecular ink synthesis method. These strategies, which include the incorporation of doping and alloying agents, pre- and post-conditioning treatments and the deposition of passivation

interlayers, have led to incremental progress when applied to absorbers prepared using solution precursors, achieving PCEs of up to 15%^{153,154}. Among the doping and alloying agents, light alkali, Ag, Cd and Ge have proven to be the most successful to date^{27,31–33,143,155–157}. The introduction of M^{3+} additives in Ag-alloyed kesterite for the passivation of grain boundaries such as In, Ga and Pd has enabled improved PCEs, going beyond 14%^{34,158}. Finally, the implementation of passivation layers at the front and back interfaces has enabled efficiencies close to 15%^{35,36}.

These strategies not only mitigate efficiency-limiting defects, but also pave the way for future advancements in the field, including the potential to drive continued improvements in device efficiency, homogeneity and reproducibility¹⁵⁹. Reaching this milestone requires a multifaceted approach, as many aspects are necessary for further development in kesterite solar cells. Kesterite solar cells continue to exhibit an important open-circuit voltage (V_{OC}) loss, with a typical deficit of 400 mV. Grain boundary recombination analysis indicates that up to 126 mV of V_{OC} losses can be attributed to recombination at grain boundaries¹⁶⁰. Device-level studies based on current, voltage, temperature and intensity suggest that up to 350 mV of losses may originate at the vicinity of the absorber/electron transport layer junction. In the latter case, losses associated with grain boundaries and bulk recombination cannot be separated¹⁶¹. To further close the V_{OC} deficit, key directions should aim to further improve the quality of CZTSSe absorbers near the junction and at the junction itself (for example, by tailoring the nucleation and grain growth stages through targeted solution additives (solvents, anionic species and sulfur sources) and employing high-temperature selenization under innovative reactive atmospheres (Se_x , SeS_2 and so on) to induce rapid crystallization while preventing the formation of intermediate phases). In parallel, extrinsic-element innovations (for example, advanced iso- and heterovalent co-alloying, introduced via the ink, which can assist in optimizing the reaction pathway) will be crucial. Strategic post-deposition chemical and thermal treatments can enable the spatial redistribution of elements, assisting the formation of compositional gradients, or promoting the formation of beneficial 2D surface phases and/or ordered vacancy compounds. Lastly, further innovation in device architecture is needed, including the development of alternatives to p(CZTSSe)–n(CdS) structures and the integration of advanced electron and hole transport layers with wide-bandgap, band and lattice matching with the kesterite absorbers, offering promising routes to enhancing carrier selectivity, minimizing interfacial recombination and exploiting the full optoelectronic potential of kesterite absorbers. By implementing these advanced-synthesis, defect-passivation, extrinsic-element and device-engineering strategies, future molecular ink kesterite photovoltaics can transcend current performance bottlenecks and fulfil their potential as high-efficiency, Earth-abundant thin-film solar absorbers.

Conclusions

This Review provides relevant insights into how precursor selection and processing conditions influence the formation pathway, defect formation and device efficiency in complex, multinary semiconductors such as kesterite. We discuss the role of solution chemistry in molecular ink approaches, highlighting how precise control of the formation pathway has enabled kesterite solar cells to surpass the critical 15% efficiency milestone. Molecular inks offer a scalable, reproducible and eco-friendly method suitable for industrial applications, positioning kesterite solar cells as a promising contender among emerging photovoltaics technologies. We discuss a clear pathway towards high-efficiency kesterite solar cells and present a solid framework for the synthesis of multinary semiconductors.

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Author contributions

A.J.-A., Y.G., J.L., H.X., X.H. and E.S. conceived of the article idea. A.J.-A. and Y.G. performed the literature search and curated the data. A.J.-A. wrote the original draft of the manuscript. A.J.-A., Y.G., I.C., O.E.K., J.L., K.S., Z.J.L.-K., S.G., H.X., A.P.-R., H.X. and E.S. reviewed and edited the manuscript. A.J.-A., Y.G., I.C., O.E.K. and S.G. visualized the results. J.L., K.S., Z.J.L.-K., S.G., X.H. and E.S. supervised the project. A.P.-R. and E.S. acquired funding. E.S. managed the project.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Alex Jimenez-Arguijo, Yuancai Gong, Xiaojing Hao or Edgardo Saucedo.

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